WORKED SOLUTION

STEP III 2014, Question 6

Pure • Analysis

STEP III 2014, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q6

The key idea

A function that starts at zero with zero slope, and whose graph bends upwards, must rise above zero. Integrating the given fact twice turns that picture into a proof. Each inequality in the question is then a matter of moving everything to one side to get a function f with $f(0) = f'(0) = 0$, and checking that f'' is positive. The real work is choosing the right arrangement, so that f'' comes out simple.

Solution

The opening result

Suppose $f''(t) > 0$ for $0 < t < x_0$ and $f(0) = f'(0) = 0$. Fix x with $0 < x < x_0$.

Take $h = f''$. Since $f''(t) > 0$ for $0 < t < x$, the given result gives

$$f'(x) = f'(x) - f'(0) = \int_0^x f''(t)\,\mathrm{d}t > 0.$$

This holds for every x in $(0, x_0)$, so $f'(t) > 0$ for $0 < t < x_0$.

Now apply the given result again, this time with $h = f'$. For $0 < x < x_0$ we have $f'(t) > 0$ for $0 < t < x$, so

$$f(x) = f(x) - f(0) = \int_0^x f'(t)\,\mathrm{d}t > 0.$$

So $f(t) > 0$ for $0 < t < x_0$, as required. We will use this with $x_0 = \frac{1}{2}\pi$ throughout.

Part (i)

Let $f(x) = 1 - \cos x \cosh x$. Then $f(0) = 1 - 1 = 0$, and by the product rule

$$f'(x) = \sin x \cosh x - \cos x \sinh x, \qquad f'(0) = 0 - 0 = 0.$$

Differentiating again,

$$\begin{aligned} f''(x) &= (\cos x \cosh x + \sin x \sinh x) - (-\sin x \sinh x + \cos x \cosh x) \\ &= 2 \sin x \sinh x. \end{aligned}$$

For $0 < x < \frac{1}{2}\pi$ both $\sin x > 0$ and $\sinh x > 0$, so $f''(x) > 0$. By the opening result, $f(x) > 0$, that is,

$$\cos x \cosh x < 1 \qquad \text{for } 0 < x < \tfrac{1}{2}\pi.$$

Part (ii)

Ask yourself Which function should we apply the opening result to? Using $\frac{\sin x}{x} - \frac{1}{\cosh x}$ directly looks messy.

Clear the fractions first. For $0 < x < \frac{1}{2}\pi$, the quantities x, $\sin x$, $\sinh x$ and $\cosh x$ are all positive, so multiplying through by positive quantities does not change the direction of an inequality. Then

$$\frac{1}{\cosh x} < \frac{\sin x}{x} \iff x < \sin x \cosh x, \qquad \frac{\sin x}{x} < \frac{x}{\sinh x} \iff \sin x \sinh x < x^2.$$

(Multiply the first by $x \cosh x > 0$, the second by $x \sinh x > 0$.) The products $\sin x \cosh x$ and $\sin x \sinh x$ differentiate neatly, which is why this is the right form.

Left-hand inequality. Let $g(x) = \sin x \cosh x - x$. Then $g(0) = 0$ and

$$g'(x) = \cos x \cosh x + \sin x \sinh x - 1, \qquad g'(0) = 1 + 0 - 1 = 0.$$

Differentiating again, the $\sin x \cosh x$ terms cancel:

$$\begin{aligned} g''(x) &= (-\sin x \cosh x + \cos x \sinh x) + (\cos x \sinh x + \sin x \cosh x) \\ &= 2 \cos x \sinh x, \end{aligned}$$

which is positive for $0 < x < \frac{1}{2}\pi$, because $\cos x > 0$ and $\sinh x > 0$ there. (This is where the upper limit $\frac{1}{2}\pi$ matters.) By the opening result, $g(x) > 0$, so $x < \sin x \cosh x$ and hence $\frac{1}{\cosh x} < \frac{\sin x}{x}$.

Right-hand inequality. Let $k(x) = x^2 - \sin x \sinh x$. Then $k(0) = 0$ and

$$k'(x) = 2x - \cos x \sinh x - \sin x \cosh x, \qquad k'(0) = 0.$$

Differentiating again, the $\sin x \sinh x$ terms cancel:

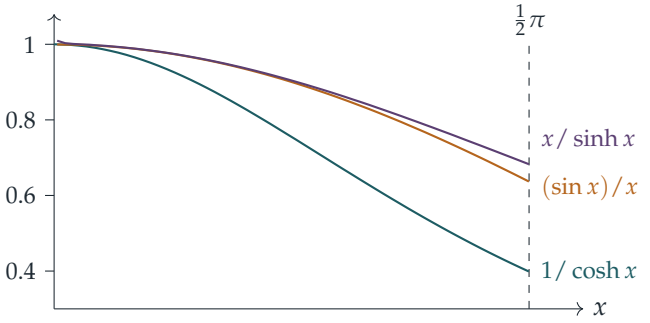
$$\begin{aligned} k''(x) &= 2 - (-\sin x \sinh x + \cos x \cosh x) - (\cos x \cosh x + \sin x \sinh x) \\ &= 2 - 2 \cos x \cosh x. \end{aligned}$$

By part (i), $\cos x \cosh x < 1$ for $0 < x < \frac{1}{2}\pi$, so $k''(x) > 0$ there. By the opening result, $k(x) > 0$, so $\sin x \sinh x < x^2$ and hence $\frac{\sin x}{x} < \frac{x}{\sinh x}$.

Putting the two together, for $0 < x < \frac{1}{2}\pi$,

$$\frac{1}{\cosh x} < \frac{\sin x}{x} < \frac{x}{\sinh x},$$

as required.



Checks

- Series: $\cos x \cosh x = 1 - \frac{1}{6}x^4 + \cdots$, so it does dip just below 1 for small x, consistent with (i). Similarly $\frac{1}{\cosh x} \approx 1 - \frac{1}{2}x^2$, $\frac{\sin x}{x} \approx 1 - \frac{1}{6}x^2$ and $\frac{x}{\sinh x} \approx 1 - \frac{1}{6}x^2$, so the left inequality is clear for small x, while the right one is decided by the next terms (the x^4 coefficients are $\frac{1}{120}$ for $\frac{\sin x}{x}$ and the larger $\frac{7}{360}$ for $\frac{x}{\sinh x}$).
- At $x = 1$: $\cos 1 \cosh 1 \approx 0.834 < 1$, and $\frac{1}{\cosh 1} \approx 0.648 < \sin 1 \approx 0.841 < \frac{1}{\sinh 1} \approx 0.851$.
- Near $x = \frac{1}{2}\pi$, at $x = 1.57$: the three values are about 0.399, 0.637 and 0.683, and $\cos x \cosh x \approx 0.002$.

Common slips

- Applying the opening result to $\frac{\sin x}{x} - \frac{1}{\cosh x}$ itself: this is not even defined at 0, and its derivatives are unpleasant. Clear denominators first, checking that every factor you multiply by is positive.
- Forgetting to check $f'(0) = 0$ as well as $f(0) = 0$; without it the opening result does not apply.
- Not noticing that the second inequality needs part (i): $k'' = 2 - 2\cos x \cosh x$ is positive precisely because of (i).

Beyond STEP

The opening result is a form of the mean value theorem argument: a function with positive derivative is increasing, applied twice. It says that a convex function lies

above its tangent line, here the tangent $y = 0$ at the origin, and this "compare second derivatives" technique is a standard way to prove inequalities between elementary functions. Abbott's *Understanding Analysis* develops the mean value theorem and its consequences carefully.

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